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C_C2_23

AI LAB

Practical 3

To implement find-s and candidate elimination algorithm on the enjoy sports dataset and compare the generated hypotheses. 1) Load Dataset 2) Perform EDA 3) And Apply Find-S algorithm.

```
In [1]: import pandas as pd
import numpy as np
```

```
In [2]: df = pd.read_csv("enjoy.csv")
```

```
In [4]: df.head()
```

```
Out[4]:
```

	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
0	Sunny	Warm	Normal	Strong	Warm	Same	Yes
1	Sunny	Warm	High	Strong	Warm	Same	Yes
2	Rainy	Cold	High	Strong	Warm	Change	No
3	Sunny	Warm	High	Strong	Cool	Change	Yes
4	Sunny	Cold	Normal	Weak	Warm	Same	No

```
In [8]: X = df.iloc[:, :-1].values
y = df.iloc[:, -1].values
```

```
In [9]: import numpy as np
def find_s(X,y):
    hypothesis = None
    print("Initial Hypothesis :", hypothesis)
    for i in range(len(X)):
        if(y[i] == "Yes"):
            if hypothesis is None:
                hypothesis = X[i].copy()
            else:
                for j in range(len(hypothesis)):
                    if hypothesis[j] != X[i][j]:
                        hypothesis[j] = "?"
        print(f"\nAfter Training Example {i+1}")
        print(hypothesis)
    return hypothesis
final_hypothesis = find_s(X,y)
```

Initial Hypothesis : None

After Training Example 1

['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']

After Training Example 2

['Sunny' 'Warm' '?' 'Strong' 'Warm' 'Same']

After Training Example 3

['Sunny' 'Warm' '?' 'Strong' 'Warm' 'Same']

After Training Example 4

['Sunny' 'Warm' '?' 'Strong' '?' '?']

After Training Example 5

['Sunny' 'Warm' '?' 'Strong' '?' '?']

After Training Example 6

['Sunny' 'Warm' '?' 'Strong' '?' '?']

After Training Example 7

['Sunny' 'Warm' '?' '?' '?' '?']

After Training Example 8

['Sunny' 'Warm' '?' '?' '?' '?']

After Training Example 9

['Sunny' 'Warm' '?' '?' '?' '?']

After Training Example 10

['Sunny' 'Warm' '?' '?' '?' '?']

Candidate Elimination Algorithm

```
In [13]: import copy

def candidate_elimination(concepts, target):
    specific = concepts[0].copy()
    general = [["?" for i in range(len(specific))]]

    print("\nInitial S:", specific)
    print("\nInitial G:", general)

    for i, h in enumerate(concepts):
        if target[i] == "Yes":
            for x in range(len(specific)):
                if h[x] != specific[x]:
                    specific[x] = "?"
        else:
            for x in range(len(specific)):
                if h[x] != specific[x]:
                    general.append(
                        ["?" if j != x else specific[j] for j in range(len(specific))
                    ]
                )
```

```
    print("\nExample", i + 1)
    print("S =", specefic)
    print("G =", general)

    return specefic, general
```

```
S, G = candidate_elimination(X, y)
```

```
print("\nFinal Specefic Boundary")
print(S)
```

```
print("\nFinal General Boundary")
for g in G:
    print(g)
```


